Stay Ahead

BANNARI AMMAN INSTITUTE OF TECHNOLOGY

An Autonomous Institution Affiliated to Anna University - Chennai • Approved by AICTE • Accredited by NAAC with "A+" Grade

SATHYAMANGALAM - 638401 ERODE DISTRICT TAMILNADU INDIA

Ph : 04295-226000/221289 Fax : 04295-226666 E-mail : stayahead@bitsathy.ac.in Web : www.bitsathy.ac.in

Quantitative Aptitude

Profit and Loss

BASIC CONCEPTS

Cost Price (C.P): Price at which one buys anything.

Selling Price (S.P): Price at which one sells anything

Profit/Loss: Difference between the Selling Price(SP) and Cost Price(CP) If the selling price is higher it is a **profit** and if the cost price is higher it is a **loss**.

Profit = SP-CP
Loss = CP - SP

FORMULAE

Profit = Selling Price - Cost Price if SP > CP
Loss = Cost Price - Selling Price if CP > SP

Profit % = (Profit / Cost Price) * 100
Loss % = (Loss / Cost Price) * 100

Selling Price = ((100 + Profit %) / 100) x Cost Price
Selling Price = ((100 - Loss %) / 100) x Cost Price
Cost Price = ((100 + Profit %) / 100) x Selling Price
Cost Price = ((100 - Loss %) / 100) x Selling Price

Profit%: **(Profit / Cost price) * 100**
Loss %: **(Loss / Cost price) * 100**

Marked price: This is basically labelled by the shopkeeper to offer a discount to the customers in such a way that,

Discount = Marked price - Selling price
Discount % = (Discount/Marked price) * 100

* Important Formulas of Profit & Loss

1. Profit = S.P. − C.P.
2. Loss = C.P. − S.P.
3. S.P. = Profit + C.P.
4. S.P. = C.P. − loss
5. C.P. = S.P. − Profit
6. C.P = Loss + S.P
7. $\text{Profit \%} = \frac{\text{Profit} \times 100}{\text{C.P}}$
8. $\text{Loss \%} = \frac{\text{Loss} \times 100}{\text{C.P}}$
9. $\text{S.P.} = \text{C.P.}\left[1 + \frac{\text{Profit}}{100}\right]$
10. $\text{C.P.} = \frac{\text{S.P.}}{\left[1 + \frac{\text{Profit}}{100}\right]}$
11. $\text{S.P.} = \text{C.P.}\left[1 - \frac{\text{loss}}{100}\right]$
12. $\text{C.P.} = \frac{\text{S.P.}}{1 - \frac{\text{loss}}{100}}$

* C.P. = Cost Price
* S.P. = Selling Price

1. If selling price is doubled, the profit triples. Find the profit percent ?

A) 100%

B) 200%

C) 300%

D) 400%

Ans: **A) 100%**

2. A trader mixes 26 kg of rice at Rs. 20 per kg with 30 kg of rice of other variety at Rs. 36 per kg and sells the mixture at Rs. 30 per kg. His profit percent is:

A) No profit, no loss

B) 5%

C) 8%

D) 10%

Ans: **B) 5%**

3. The cost price of 30 articles is equal to the selling price of 25 articles. What is the profit percentage in this transaction?

A) 20%

B) 18%

C) 15%

D) 13%

Ans: **A) 20%**

4. If the cost price of 12 pens is equal to the selling price of 8 pens, the gain percent is ?

A) 12%

B) 30%

C) 50%

D) 60%

Ans: **C) 50%**

5. A man sold 18 cots for Rs. 16,800, gaining thereby the cost price of 3 cots. The cost price of a cot is :

A) 800

B) 900

C) 600

D) 700

Ans: **A) 800**

6. A man bought oranges at 5 for a rupee. How many for a rupee must he sell to gain 25%?

A) 3

B) 6

C) 4

D) 8

Ans: **C) 4**

7. A shopkeeper bought a watch for Rs. 1200. At what price should he sell the watch to earn a profit of 25% ?

A) Rs. 1400

B) Rs. 1450

C) Rs. 1500

D) Rs. 1550

Ans: **C) Rs. 1500**

8. If the cost price of 120 mangoes is equal to the selling price of 100 mangoes, what is the profit percent in this transaction?

A) 15%

B) 17%

C) 20%

D) 21%

Ans: **C) 20%**

9. Two articles are sold at Rs.199 each such that a profit of 10% is made on the first while a loss of 10% is incurred on the other. What would be the net profit/loss on the two transactions?

A) Loss of 1%

B) Loss of 10%

C) Gain of 1%

D) Gain of 10%

Ans: **A) Loss of 1%**

10. The profit percent gained if an object is sold at Rs. 464 is equals to the loss percent when the object is sold at Rs. 436. Find the cost price of this object?

A) 250

B) 400

C) 450

D) 500

Ans: **C) 450**

11. The owner of the cell phone shop charges 23% more than the cost price. If a customer paid 7011 for a cell phone, find the cost price of the cell phone.

A) 5860

B) 5220

C) 5700

D) 6750

Ans: **C) 5700**

12. By selling 45 lemons for Rs 40, a man loses 20%. How many should he sell for Rs 24 to gain 20% in the transaction ?

A) 16

B) 18

C) 20

D) 22

Ans: **B) 18**

13. By selling an object for Rs. 900, a man earns a profit that is double the loss incurred when the object is sold for Rs. 450. If he sells it at a profit of 20%, find the selling price of this object.

A) 620

B) 618

C) 554

D) None of these

Ans: **D) None of these**

14. By selling a property for Rs. 45000 a person incurs a loss of 10%. Find the selling price to gain the profit of 15 %?

A) 55000

B) 60000

C) 57500

D) 58000

Ans: **C) 57500**

15. A person purchases some article at the rate of 4 articles for rupees 5 and sells them at the rate of 5 articles for rupees 4. Find the total profit or loss in %.

A) 22% profit

B) 20% loss

C) 40% profit

D) 36% loss

Ans: **B) 20% loss**

BANNARI AMMAN INSTITUTE OF TECHNOLOGY
An Autonomous Institution Affiliated to Anna University - Chennai • Approved by AICTE • Accredited by NAAC with "A+" Grade
SATHYAMANGALAM - 638401 ERODE DISTRICT TAMILNADU INDIA
Ph : 04295-226000/221289 Fax : 04295-226666 E-mail : stayahead@bitsathy.ac.in Web : www.bitsathy.ac.in

PERCENTAGES

1. Find the 30% of a number whose 25% is 30% of 400.

A. 15

B. 12

C. 9

D. 18

ANS : 9

2. 40% of x = 10% of 300 + 75% of 120. Find x.

A. 150
B. 120
C. 300
D. 280

ANS : 300

3. Ajay saves 25% of his income. If his income is Rs.61,200/annum, find his monthly saving.

A. 1500
B. 1275
C. 3000
D. 2800

ANS : 1275

4. What percentage increase when 288 is increased to 320?

A. 10%
B. 11.11%
C. 9.09%
D. 15%

ANS : 11.11%

5. In Gita's birthday, 75% of her friends came, 11 friends did not come and 7 friends are on their way to the party. Find the total number of friends she has.

A. 82
B. 92
C. 72
D. 80

ANS : 72

6. A Stationery seller had some Pens, Sharpeners, Erasers & Pencils. He sells 65% of the total units and still has 175 units. Originally, he had:

A. 10
B. 16 2/3
C. 17 1/3
D. 20

ANS : 500 units

7. If the annual income of X is 20% more than that of Y, then the income of Y is less than that of X by p%. What is the value of p?

A. 10
B. 16 2/3
C. 17 1/3
D. 20

ANS : 16 2/3

8. In ABC College, 65% of students are less than 20 years of age. The number of students more than 20 years of age is 2/3 of number of students of 20 years of age which is 42. What is the total number of students in the ABC College?

A. 75
B. 90
C. 130
D. 200

ANS : 200

9. Two friends, Arun & Ajay had some candies each. One of them had 15 candies more than the other. The candies with Arun was 60% of the total candies with them. How many candies did each have?

A. 40, 25

B. 47, 32

C. 45, 30

D. 49, 34

ANS : 45, 20

10. 3/8 percent of the value of the car is 3500 rupees, then what is the price of the car?

A. Rs.933333
B. Rs.830000
C. Rs.900000
D. Rs.933330

ANS :
Rs.933333

11. A is 10% more than 1250. B is 10% less than A. then, Find the value of B?

A. 225
B. 320
C. 400
D. 200

ANS : 225

12. The length of a rectangle is increased by 10% and breadth is decreased by 10%. Then the area of the new rectangle is

A. Neither increased nor decreased

B. Increased by 1%

C. Decreased by 1%

D. Decreased by 10%

ANS : Decreased by 10%

13. A shopkeeper bought 600 oranges and 400 bananas. He found 15% of oranges and 8% of bananas were rotten. Find the percentage of fruits in good condition?

A. 66

B. 88

C. 77

D. 55

ANS : 88

14. A candidate who gets 25% marks fails by 10 marks but another candidate who gets 36% marks gets 12 more than the passing marks. Find the maximum marks.

A. 300

B. 200

C. 100

D. 150

ANS : 200

15. In a class of 125 students, if the ratio of boys and girls in a class is 3:2. If 24% of the boys and 20% of the girls are interested in dance, then find the % of students who are all not interested in dance?

A. 72.5 %

B. 75.8 %

C. 77.6 %

D. 65.6 %

ANS : 77.6 %

Stay Ahead

BANNARI AMMAN INSTITUTE OF TECHNOLOGY

An Autonomous Institution Affiliated to Anna University - Chennai • Approved by AICTE • Accredited by NAAC with "A+" Grade

SATHYAMANGALAM - 638401 ERODE DISTRICT TAMILNADU INDIA

Ph : 04295-226000/221289 Fax : 04295-226666 E-mail : stayahead@bitsathy.ac.in Web : www.bitsathy.ac.in

AGES

1. The sum of the present ages of A and B is 40 years. A is 4 years older than B. What are their present ages?

- A) 18, 22
- B) 20, 20
- C) 25, 15
- D) 22, 18

Answer D) 22, 18

2. A's age is three times B's age. If the sum of their ages is 36, what is A's age?

- A) 27
- B) 18
- C) 24
- D) 12

Answer A) 27

3. The present age of a father is 4 times the present age of his son. After 5 years, the father's age will be three times the son's age. What is the present age of the son?

- A) 5 years
- B) 10 years
- C) 6 years
- D) 8 years

Answer A) 5 years

4. 5 years ago, a man's age was 6 times the age of his son. After 5 years, the man's age will be 4 times his son's age. What is the present age of the son?

- A) 15 years
- B) 10 years
- C) 5 years
- D) 8 years

Answer B) 10 years

5. A man is 24 years older than his son. In 4 years, he will be twice as old as his son. What is the present age of the son?

- A) 10 years
- B) 12 years
- C) 8 years
- D) 16 years

Answer A) 10 years

6. The age of a father is 5 years more than twice the age of his son. If the sum of their ages is 45 years, what is the present age of the son?

- A) 10 years
- B) 12 years
- C) 15 years
- D) 20 years

Answer A) 10 years

7. A father is twice as old as his son. After 10 years, the father's age will be 10 years more than 3 times the son's age. What is the present age of the father?

- A) 40 years
- B) 50 years
- C) 30 years
- D) 60 years

Answer A) 40 years

8. A person is 6 years older than his sister. In 4 years, he will be twice as old as she will be. What is the present age of the person?

- A) 8 years
- B) 12 years
- C) 14 years
- D) 10 years

Answer D) 10 years

9. The sum of the present ages of a mother and her daughter is 70 years. The mother is 4 times as old as the daughter. What is the daughter's present age?

- A) 10 years
- B) 15 years
- C) 20 years
- D) 18 years

Answer C) 14 years

10. The present age of a woman is 4 times the age of her daughter. After 6 years, the woman will be three times the age of her daughter. What is the present age of the daughter?

- A) 6 years
- B) 4 years
- C) 8 years
- D) 5 years

Answer B) 4 years

11. The sum of the ages of a man and his wife is 70 years. The wife is 10 years younger than the man. What are their ages?

- A) 40, 30
- B) 45, 25
- C) 50, 20
- D) 55, 15

Answer A) 40, 30

12. The sum of the ages of a mother and her daughter is 48 years. The mother is 6 times as old as the daughter. What is the daughter's present age?

- A) 6 years
- B) 8 years
- C) 9 years
- D) 12 years

Answer B) 8 years

13. A man’s age is 10 more than 3 times his daughter’s age. If the man’s age is 58 years, what is the daughter’s age?

- A) 12 years
- B) 13 years
- C) 14 years
- D) 15 years

Answer C) 16 years

14. A father is 10 years older than the mother of his child. The sum of their ages is 100 years. What is the present age of the father?

- A) 50 years
- B) 55 years
- C) 60 years
- D) 45 years

Answer B) 55 years

15. The sum of the present ages of a father and his son is 56 years. The father is 6 times as old as the son. What are their ages?

- A) 42, 14
- B) 48, 8
- C) 54, 2
- D) 30, 26

Answer B) 48, 8

BANNARI AMMAN INSTITUTE OF TECHNOLOGY
An Autonomous Institution Affiliated to Anna University - Chennai • Approved by AICTE • Accredited by NAAC with "A+" Grade
SATHYAMANGALAM - 638401 ERODE DISTRICT TAMILNADU INDIA
Ph : 04295-226000/221289 Fax : 04295-226666 E-mail : stayahead@bitsathy.ac.in Web : www.bitsathy.ac.in

Calendar

1.What day was 26 January 1950?

A) Thursday
B) Friday
C) Saturday
D) Sunday

Ans: **B)Friday**

2. What day will be 1 January 2030?

A) Monday
B) Tuesday
C) Wednesday
D) Thursday

Ans: **C)Wednesday**

3. What day was 2 October 1869?

A) Friday
B) Saturday
C) Sunday
D) Monday

Ans: **C)Sunday**

4. What day will be 29 February 2028?

A) Monday
B) Tuesday
C) Wednesday
D) Thursday

Ans: **C)Wednesday**

5. What day was 1 March 2000?

A) Tuesday
B) Wednesday
C) Thursday
D) Friday

Ans: **B)Wednesday**

Thank you